Diagnostic testing for Ebola disease and Marburg virus disease

Interim guidance, 9 July 2026

World Health Organization

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Declaration of interests

All external experts who were involved in the development and review of the interim guidance completed a WHO declaration of interest to disclose potential conflicts of interest that might affect, or might reasonably be perceived to affect, their objectivity and independence concerning the subject. WHO reviewed each of the declarations and concluded that none could give rise to a potential or reasonably perceived conflict of interest related to the subjects reviewed.

Abbreviations

BDBV	Bundibugyo virus
BOMV	Bombali virus
BVD	Bundibugyo virus disease
Cq	Cycle quantification
Ct	Cycle threshold
EBOD	Ebola disease
EBOV	Ebola virus
EDTA	Ethylenediaminetetraacetic acid
EVD	Ebola virus disease
MARV	Marburg virus
MVD	Marburg virus disease
NAAT	Nucleic acid amplification testing
PPE	Personal protective equipment
RAVV	Ravn virus
RNA	Ribonucleic acid
RT-PCR	Reverse transcription polymerase chain reaction
SUDV	Sudan virus
SVD	Sudan virus disease
TAFV	Taï Forest virus
TVD	Taï Forest virus disease
VHF	Viral haemorrhagic fever
WHO	World Health Organization

Key points

- The goal of diagnostic testing for Ebola disease (EBOD), caused by orthoebolaviruses including Ebola, Sudan, Bundibugyo, and Taï Forest viruses, and Marburg virus disease (MVD), caused by orthomarburgviruses including Marburg and Ravn viruses, is to confirm cases to provide timely and appropriate care, implement appropriate infection prevention and control measures, and interrupt disease transmission, in the context of surveillance, outbreak investigation and public health response.

- All individuals meeting the case definition for suspected EBOD or MVD should be tested.

- Laboratory testing of asymptomatic individuals, including asymptomatic contacts, is not recommended, including for travel or point-of-entry screening purposes.

- The recommended specimen type to collect for testing for orthoebolaviruses and orthomarburgviruses is whole blood (EDTA blood) for living patients, and oral swab for deceased individuals.

- The preferred assay type to confirm an infection with an orthoebolavirus or an orthomarburgvirus is nucleic acid amplification testing (NAAT).

- For any new outbreak, if cases are not known to be epidemiologically linked to confirmed cases, and in the case of importation or cross-border transmission, testing should enable confirmation of the virus genus (*Orthoebolavirus* or *Orthomarburgvirus*), and the virus species involved.

- For orthoebolavirus infections, species differentiation (*Orthoebolavirus bundibugyoense, Orthoebolavirus sudanense, Orthoebolavirus taiense* or *Orthoebolavirus zairense*) is critical as countermeasures differ according to the orthoebolavirus species involved.

- Whole or partial genomic sequencing can be used to complement NAAT diagnostic testing where appropriate. It can be used to characterize viruses and complement epidemiologic investigations by providing insights into transmission chains and the origin of the outbreak.

- If a suspected case of EBOD or MVD in a living patient tests negative and the initial blood sample was collected within 72 hours of symptom onset, a second test should be performed on a new blood sample, collected more than 72 hours after symptom onset.

- Differential diagnosis should always be considered and testing conducted either sequentially or in parallel as appropriate, according to the clinical symptoms and the epidemiological context.

- All manipulations in laboratory settings of specimens originating from suspected, probable or confirmed cases of EBOD or MVD should be conducted by adequately trained and competent personnel with appropriate biosafety measures according to a risk-based approach.

- States Parties to the International Health Regulations are reminded of their obligations to use the decision instrument in Annex 2 of the International Health Regulations (2005) and notify the World Health Organization (WHO) accordingly of positive laboratory results from a suspected case of EBOD or MVD.

- Member States are encouraged to share genetic sequence data in publicly accessible databases.

1. Introduction

1.1 Background

The scope of this guidance is to provide interim recommendations regarding the diagnosis and laboratory testing of Ebola disease (EBOD) and Marburg virus disease (MVD). The objectives are to describe laboratory testing for EBOD and MVD, and to propose diagnostic strategies.

This guidance constitutes an update to the *Diagnostic testing for Ebola and Marburg virus diseases: interim guidance, 20 December 2024 (1)*. This update was necessary in order to incorporate the latest understanding and best practices for the diagnosis of MVD and EBOD, in the context of the 2026 Bundibugyo virus disease (BVD) outbreak. It strengthens guidance on species identification, and the selection and use of diagnostic assays, notably clarifying the role of NAAT and serology, as well as genomic sequencing.

The guidance further expands recommendations on specimen collection and handling, including storage, biological risk management, and the use of complementary diagnostic methods. Greater emphasis is placed on differential diagnosis and coinfections, as well as on the organization of national testing strategies, including decentralized testing capacity.

In addition, the document includes enhanced guidance on reporting, data use and sharing of samples and genomic data, with consideration for equity, ethical frameworks and international collaboration.

1.2 Target audience

This document provides interim guidance for laboratories, clinicians, health workers, public health officials, outbreak response teams, and other stakeholders involved in the diagnosis and care of patients with suspected or confirmed EBOD or MVD, from specimen collection through to diagnostic analysis and reporting.

1.3 Methodology

This guidance was developed based on the latest available evidence on EBOD, MVD and diagnostics, and on the World Health Organization (WHO) normative guidance and technical documents related to diagnostic testing for EBOD and MVD, including the *Diagnostic testing for Ebola and Marburg virus diseases: interim guidance, 20 December 2024 (1)*.

Given the 2026 Bundibugyo virus disease (BVD) outbreak and the declaration of a Public Health Emergency of International Concern, this interim guidance was updated under a fast-track emergency process to provide timely recommendations for outbreak response. The update considered evidence available up to July 2026, including published scientific literature,

operational experience from previous EBOD and MVD outbreaks, WHO technical guidance and normative documents, and information generated during the ongoing BVD outbreak response.

The revision process was coordinated by WHO and involved a multidisciplinary group of internal and external experts with expertise in laboratory diagnostics, surveillance, genomic sequencing, clinical management, infection prevention and control, biosafety, biosecurity, and outbreak response. Draft revisions were reviewed through technical consultations and document review rounds.

Proposed changes to the 2024 guidance were assessed according to the available evidence, consistency with existing WHO normative guidance, operational feasibility during outbreak response, and potential implications for patient management, surveillance and public health action. Where direct evidence was limited or unavailable, recommendations were informed by expert consensus and experience from previous filovirus outbreaks. Consensus was reached through iterative review and discussion among WHO and external experts.

Because evidence remains limited in some areas, particularly regarding diagnostic test performance for some viruses, use of antigen rapid diagnostic tests and selected operational practices, some recommendations are necessarily provisional and may be revised as additional evidence becomes available.

2. Filovirus causing human disease

The family *Filoviridae* includes the genera *Orthoebolavirus* and *Orthomarburgvirus*, which comprise single-stranded, negative-sense RNA viruses. Six *Orthoebolavirus* species have been characterized to date, each currently represented by a single virus: Bombali virus (BOMV), Bundibugyo virus (BDBV), Reston virus (RESTV), Sudan virus (SUDV), Taï Forest virus (TAFV), and Ebola virus (EBOV) *(2)*. In contrast, a single species within *Orthomarburgvirus* has been identified, which includes two viruses, Marburg virus (MARV) and Ravn virus (RAVV) *(3)*. A summary of the classification according to the International Committee on Taxonomy of Viruses is shown in Table 1 *(4)*.

BDBV, SUDV, TAFV, EBOV, MARV and RAVV have been associated with outbreaks in humans. These viruses are transmitted to humans from wild animals (zoonotic transmission) and spread in the human population through human-to-human transmission. To date, only one case of human disease with TAFV has been described. BOMV and RESTV have not been associated with human disease to date.

Table 1. Virus classification according to the International Committee on Taxonomy of Viruses, 2024[1]

Family	Genus	Species	Virus name	Disease name
Filoviridae	*Orthoebolavirus*	*Orthoebolavirus bombaliense*	Bombali virus (BOMV)	Not known to cause human disease to date
		Orthoebolavirus bundibugyoense	Bundibugyo virus (BDBV)	Bundibugyo virus disease (BVD)
		Orthoebolavirus restonense	Reston virus (RESTV)	Not known to cause human disease to date
		Orthoebolavirus sudanense	Sudan virus (SUDV)	Sudan virus disease (SVD)
		Orthoebolavirus taiense	Taï Forest virus (TAFV)	Taï Forest virus disease (TVD)
		Orthoebolavirus zairense	Ebola virus (EBOV)	Ebola virus disease (EVD)
	Orthomarburgvirus	*Orthomarburgvirus marburgense*	Marburg virus (MARV)	Marburg virus disease (MVD)
			Ravn virus (RAVV)	Marburg virus disease (MVD)

Disease caused by infection with an orthoebolavirus may be referred to by the specific disease names as described above in the table, or by the generic term "Ebola disease" (EBOD). Disease caused by infection with either of the orthomarburgviruses is referred to as Marburg virus disease (MVD). For more information on disease names, see the *International Classification of Diseases 11th Revision (5)*.

[1] Note that there has been a recent change in filovirus nomenclature, more information can be found through the International Committee on Taxonomy of Viruses *(4)*.

The incubation period for EBOD and MVD is most frequently 3–10 days but can range from 2 to 21 days *(6)*. Individuals are not considered infectious until symptom onset. Orthoebolavirus and orthomarburgvirus infections typically present with symptoms that are non-specific and can include fever, headache, lethargy, anorexia/loss of appetite, aching muscles or joints, stomach pain, difficulty swallowing, vomiting, difficulty breathing and diarrhoea. As the disease advances, patients may experience internal or external bleeding and organ failure. Standard case definitions can be found in the *Ebola and Marburg virus outbreak toolbox (7)*. Case fatality rates of previous outbreaks vary from 24% to 90% *(8, 9)* but reduce with early diagnosis, optimized supportive care and targeted therapeutics and, where applicable, vaccines *(10, 11)*.

Viruses within the *Orthoebolavirus* genus have genomes that differ by at least 30% at the nucleotide level *(12)*. Countermeasures, including vaccines and therapeutics, are specific to each *Orthoebolavirus* and *Orthomarburgvirus* species. At the time of the publication of this guidance, licensed vaccines and therapeutics are only available for *Orthoebolavirus zairense*. Some diagnostic tests are designed to detect only a single *Orthoebolavirus* species, whereas others can detect multiple species. This will influence their usability for testing in different situations and should be considered in the epidemiological context.

Given all the clinical, diagnostic and therapeutic implications, it is therefore important to characterize the causative virus to the species level to enable the deployment of appropriate diagnostic tests and countermeasures. Box 1 below gives further information on BDBV, in light of the 2026 outbreak of BVD.

Box 1. BDBV focus, in light of the 2026 outbreak of BVD

Data on EBOD caused by BDBV remain limited. Prior to the 2026 outbreak, which was declared a Public Health Emergency of International Concern on 17 May 2026, two outbreaks were reported: in Uganda (2007–2008) and the Democratic Republic of the Congo (2012), resulting in a combined total of 211 reported cases and 71 deaths *(13)*. The incubation period is estimated to range from 2 to 21 days and the case fatality ratio from approximately 32% to 41%. Its genome organization is similar to that of other orthoebolaviruses, but it shares only approximately 68% nucleotide identity with *Orthoebolavirus zairense*, which may have implications for the performance of diagnostics, vaccines and therapeutics developed primarily against Ebola virus.

3. Indications for testing and importance of differential diagnosis and coinfections

In vitro diagnostic testing for EBOD and MVD is primarily indicated for:

- any individual (living or deceased) meeting the case definition for a suspected case; and
- confirming viral clearance prior to discharge of a confirmed patient from a treatment centre. Where feasible, two consecutive negative nucleic acid amplification testing (NAAT) results from blood samples collected at least 48 hours apart are recommended.[2]

Testing may also be performed to support clinical follow-up during the acute phase of illness, after discharge, and to support research activities, including studies on medical countermeasures.

If a suspected case of EBOD or MVD in a living patient tests negative and the initial blood sample was collected within 72 hours of symptom onset, a second test should be performed using a new blood sample, collected more than 72 hours after symptom onset.

Viraemia generally appears at symptom onset, and the likelihood of detecting viral RNA before symptom onset is very low, even using NAAT. A negative result in an asymptomatic person does not rule out infection or the possibility of developing disease during the incubation period, and may lead to false reassurance, inappropriate modification of public health measures, unnecessary use of laboratory resources, and adverse social or economic consequences. Laboratory testing of asymptomatic individuals, including for travel-related requests or point-of-entry screening, is therefore not recommended. Contacts should be monitored for the development of symptoms for 21 days following their most recent exposure and tested only if they meet the case definition.

Given the non-specific clinical presentation of EBOD and MVD, it may not be possible to distinguish them clinically from other infectious diseases. Differential diagnoses and investigations for coinfections should therefore be guided by the epidemiological context and diseases endemic to the affected setting. These may include malaria; bacterial infections causing sepsis (such as leptospirosis, rickettsiosis and typhoid fever); and other viral haemorrhagic

[2] Bodily fluids other than blood, for example semen, may still contain detectable RNA after it is no longer detected in the blood. Patients with no RNA detected in the blood should be discharged, provided they are clinically well. In exceptional circumstances, such as limited treatment capacity or operational constraints, a single negative test result may be considered sufficient for discharge based on clinical assessment, provided that symptoms have resolved and appropriate follow-up measures are in place. Such decisions should be made on a case-by-case basis by the attending clinician. All recovered patients should be able to access a care programme that will support them after discharge, including testing of relevant body fluids and counselling on safer sex practices and hygiene.

fevers (VHFs), including dengue, yellow fever, Rift Valley fever, Lassa fever and Crimean-Congo haemorrhagic fever.

Where initial differential diagnostic investigations are negative, additional testing using advanced laboratory approaches, such as metagenomic sequencing, may be considered to support pathogen identification. If such capacity is not available in-country, referral to appropriately equipped international reference laboratories may be considered.

In patients with confirmed orthoebolavirus or orthomarburgvirus infection, repeat NAAT testing during the course of illness may support clinical management; however, cycle quantification (Cq), or cycle threshold (Ct), values (i.e., wording may vary depending on the NAAT platform used) should be interpreted with significant caution. Investigations for coinfections should also be undertaken, as appropriate to the epidemiological setting, in order to ensure optimal clinical management.

Serological testing may be used to identify individuals with prior exposure to an *Orthoebolavirus* or *Orthomarburgvirus*, including recent past infection, to support epidemiological investigations and facilitate access to appropriate follow-up and survivor care where indicated.

Specific recommendations on clinical testing for patient management are available in the *WHO guidelines for the clinical management of filovirus disease (14)*.

4. Specimen types, collection, transportation and storage considerations

4.1 Specimen types

Required specimen types vary depending on the analyses to be conducted at the laboratory. The recommended specimen types for laboratory confirmation of orthoebolavirus and orthomarburgvirus infections are described in Table 2.

Table 2. Summary of indications for testing, preferred assay types for each indication and sample types to be collected

Testing purpose	Preferred assay type	Preferred specimen type to collect
Diagnostic or follow up testing – living patient	NAAT	EDTA-blood *Capillary blood collected by heel-prick may be considered in infants when venous blood collection is not feasible, although reduced sensitivity should be anticipated (15).* *Dry blood swabs may be used but may lead to a significant loss of sensitivity.* *Oral swabbing is not recommended for diagnosis in living patients because viral RNA concentrations are generally lower than in blood specimens, resulting in reduced diagnostic sensitivity.*
Diagnostic testing – deceased individual	NAAT	Oral swab, placed in viral transport medium *The preferred swab type is nylon flocked swabs. Swabbing should aim to collect oral fluid from the inner cheeks and/or gum (crevicular gingival fluids).* *Swabbing of the nasopharynx is not recommended as the likelihood of detecting virus is not well understood.* *Dry swabs may be used if the lab is equipped to resuspend them in nuclease-free water.*
Identify previous exposure – asymptomatic or convalescent individual who may have been previously exposed within the current outbreak context or in the past	Serology	Whole blood collected in a tube without anticoagulant *Serum is preferable but EDTA-plasma can be used.*
Patient follow-up during the acute phase of illness, post discharge or for research purposes	Multiple	Depending on clinical parameters and epidemiological investigations, EDTA-blood/plasma or serum, or other specimen types such as cerebrospinal fluid, urine, breast milk, semen, amniotic fluid, placenta etc. may be collected

4.2 Specimen collection, storage considerations and transportation

Specimens should be collected in accordance with the indication for testing and the type of diagnostic test to be conducted. All samples collected should be regarded as potentially infectious and handled with extreme caution (see section on biological risk management of this guidance document), and sample collection should only be performed by those trained in the collection of filovirus samples. More information can be found in the interim guidance on *How to safely collect samples by phlebotomy from patients suspected to be infected with Ebola or Marburg (16)* and in the *Interim guideline: how to safely collect oral swabs (saliva) from deceased patients suspected to be infected with Ebola (17)*.

Specimens should be transferred to the laboratory accompanied by the laboratory information sheets, including appropriate chain of custody forms to track samples, and a copy of the case investigation form. All forms need to be packed so that they are separated from the secondary container. An example of a case investigation form can be found in the *Ebola and Marburg virus outbreak toolbox (7)*.

Correct handling and storage of samples during transportation and upon arrival at the laboratory is essential for biosafety and for accurate diagnostic testing. More information on sample storage conditions can be found in Table 3 below.

Table 3. Considerations for specimen/sample storage conditions

Specimen/ sample type	Storage conditions
EDTA-blood and whole blood	≤ 24 hours: ambient temperature (up to 25 °C)
	1–7 days: 2–8 °C
	Once EDTA-blood and/or whole blood reaches the laboratory, plasma and/or serum should be separated.
	Where prolonged storage is required for surveillance or research purposes, samples may be stored and transported on dry ice (–70 °C) (18). However, it should be noted that freezing may cause haemolysis and/or viral degradation, which could affect downstream analyses.
Plasma (from EDTA-blood) and serum	≤ 24 hours: ambient temperature (up to 25 °C)
	1–7 days: 2–8 °C
	> 7 days: –70 °C (dry ice), if not available store at –20 °C
	> 60 days from collection: –70 °C
	Note that the separation of blood into plasma and/or serum and the aliquoting of samples should only be done in an appropriately equipped laboratory and by adequately trained personnel (see biological risk management section).
	When plasma (and/or serum) separation by centrifugation is not feasible, passive separation by gravity may be applied. Before freezing (–20 °C or –70 °C), EDTA-plasma and serum samples should be aliquoted into cryogenic tubes. Freeze-thaw cycles should be avoided as this may affect sample quality.

Table 3. Considerations for sample storage conditions *(cont.)*

Specimen/ sample type	Storage conditions
Oral swabs in viral transport medium or nuclease free water	≤ 24 hours: ambient temperature
	1–7 days: 2–8 °C
	> 7 days: –70 °C; if not available, store at –20 °C
	> 60 days from collection: –70 °C
	Before freezing (–20 °C or –70 °C), viral transport medium or nuclease free water suspension from samples should be aliquoted into cryogenic tubes. Freeze-thaw cycles should be avoided as this may affect sample quality. Note that aliquoting of samples should only be done in an appropriately equipped laboratory (see biological risk management section). Dry swabs that have not been resuspended in nuclease free water should not be frozen.

Samples should be appropriately triple packaged and transported to the nearest designated orthoebolaviruses and orthomarburgviruses testing laboratory as soon as possible after collection, and preferably within 24 hours. Transport of samples should comply with national and/or international regulations, including the *UN Model Regulations Rev. 23 (2023) (19)* and any other applicable regulations depending on the mode of transport being used.

For international transport, samples from suspected, probable or confirmed cases of orthoebolavirus and orthomarburgvirus infection, including clinical samples, viral isolates and cultures should be transported as Category A, UN2814 "Infectious Substance, Affecting Humans". International shipping requires a dangerous goods declaration form and a certified shipper. For information on infectious substances shipping requirements, please see the WHO *Guidance on regulations for the transport of infectious substances 2025–2026: applicable as from 1 October 2025 (20)*.

Upon reception at the laboratory, samples (triple packaged) should be inspected for any evidence of leaks or inappropriate packaging (see biological risk management).

5. Laboratory testing methods

A unique identification number should be allocated upon arrival at the laboratory for tracking purposes. The unique identification number should be assigned to the patient sample and patient form, with all subsequent downstream products also being assigned this number, such as any aliquots of the sample and extracted nucleic acid, for example. Testing for the presence of orthoebolaviruses or orthomarburgviruses should be performed in appropriately equipped laboratories by staff trained and competent in the relevant technical and biosafety procedures (see section on biological risk management of this guidance document).

Member States are encouraged to establish, within national facilities or through collaboration with international facilities, appropriate systems for sample retention, storage and biobanking. They are also encouraged to facilitate the equitable sharing of clinical samples and derived materials (e.g. inactivated samples, extracted nucleic acids and, where appropriate, viral isolates) to support research and development of medical countermeasures. Sample retention and sharing should comply with applicable national and international regulations and be guided by ethical and legal frameworks, including biosafety and biosecurity requirements, with due consideration to equity, transparency, benefit sharing, and appropriate acknowledgement of originating laboratories and countries.

5.1 Sample inactivation for diagnostic workflow

Samples that may contain orthoebolaviruses or orthomarburgviruses should be inactivated using validated virus inactivation methods before samples can be manipulated outside the containment area (which may be a Class III biosafety cabinet or glovebox as described in the section on biological risk management). Examples of inactivation methods include incubation with 50–70% guanidine thiocyanate plus absolute ethanol (molecular biology grade, ≥ 99.8%, certified free of DNase, RNase and protease activity), heat treatment at 60 °C for 15 minutes or use of detergent. These methods should be validated regarding their efficacy and their compatibility with downstream testing of the samples.

5.2 NAAT for orthoebolaviruses and orthomarburgviruses

In the context of an outbreak caused by an orthoebolavirus or an orthomarburgvirus, the method of choice for the detection of viral RNA is NAAT. This usually refers to real time reverse transcription polymerase chain reaction (RT-PCR) but isothermal amplification or other molecular techniques may also be used if available and appropriately validated. It is recommended to use a validated assay with known analytical sensitivity and specificity in the diagnosis of EBOD and MVD due to the implications of the result for both the patient management and outbreak response.

NAAT is currently the preferred technique as it is relatively fast (<6 hours to perform), is conducted on inactivated samples and without virus propagation, lowering biological risk and infrastructure requirements. It is also the only documented method that allows for early confirmation of infection, as viral RNA can be detected as early as 48 to 72 hours after symptom onset.

Note that when using RT-PCR, a Cq or Ct value is associated with each positive sample. A Cq/Ct value is the number of amplification cycles required for the sample to be detected as positive. There is an inverse relationship between Cq/Ct values and the amount of virus RNA in a patient sample.

At the beginning of an outbreak, testing should identify the specific *Orthoebolavirus* or *Orthomarburgvirus* species causing the outbreak. Once the causative species has been confirmed, testing capacity may be scaled up using assays that differentiate only between the *Orthoebolavirus* and *Orthomarburgvirus* genera, without identifying the specific species. However, the possibility of simultaneous outbreaks involving different species should be considered. Therefore, species-level identification should be conducted for cases that are not epidemiologically linked to confirmed cases to detect the emergence of additional species.

There is no single standardized testing algorithm applicable across all settings. Diagnostic approaches should be adapted to the epidemiological context, available laboratory capacity, and testing objectives. Depending on the context, the assays available and their targets, several consecutive steps may be required, with for example the use of a pan-filovirus assay, and then an assay enabling the identification of the genus and/or of the species involved. The different types of assays that may be used are described in Table 4. Whole or partial genomic sequencing may be used, where available and appropriate, to complement NAAT diagnostic testing. Its use should be guided by public health objectives and laboratory capacity and resources (see section on genomic sequencing).

Table 4. Assay targets and purposes for diagnostic testing

Assay target	Testing purpose
Pan-filovirus NAAT	To confirm if the infection is caused by a filovirus affecting humans (*Orthoebolavirus* or *Orthomarburgvirus)*
Orthoebolavirus genus specific NAAT	To confirm if the infection is caused by any *Orthoebolavirus* species
Orthoebolavirus species specific NAAT	To confirm if the infection is caused by *Orthoebolavirus bundibugyoense*, *restonense*, *sudanense*, *taiense* or *zairense* species
Orthomarburgvirus species specific NAAT	To confirm if the infection is caused by *Orthomarburgvirus marburgense* (MARV or RAVV)

Few commercial testing kits are currently available, among which six obtained WHO Emergency Use and Assessment Listing status for testing for *Orthoebolavirus zairense*, during the 2014 to 2016 EVD outbreak response. Of these, one is pan-filovirus, one is pan-ebolavirus and four are specific for *Orthoebolavirus zairense.* While two of the products claim to detect other *Orthoebolavirus* species, evaluation of that claim was not undertaken under the Emergency Use Assessment and Listing process at that time. The list can be accessed on the WHO website *(21)*. Note that while the Emergency Use Assessment and Listing carried out up to 2016 remains valid, kits that have been commercialized since the close of the Emergency Use Assessment and Listing have not been assessed through this mechanism and may also be a viable option for testing. In the context of the 17 May 2026 declaration of a Public Health Emergency of

International Concern for the Ebola disease outbreak caused by BDBV, WHO has initiated the Emergency Use Listing[3] procedure *(22)* for relevant in vitro diagnostics. WHO will publish and update the products assessed and listed under the Emergency Use Listing as the process progresses *(23)*. At time of publication, one product was listed.

Note also that some available commercial kits are designated as research use only.

There are several primers and probe sets for RT-PCR assays for *orthoebolaviruses* and *orthomarburgviruses* that have been published and can be used for in-house development of assays in laboratories with appropriate capacities.

Before an assay is utilized to test human clinical samples within a laboratory, its appropriateness for the testing needs should be carefully considered. Its performance should also be validated or verified, as appropriate, within the laboratory, including reagents, consumables and equipment used by appropriately trained and competent staff.

5.3 Serology

Serological tests detect antibodies (IgM or IgG) produced following exposure to *orthoebolaviruses* or *orthomarburgviruses*. Serological tests are important tools in identifying individuals who have experienced prior infection or have been vaccinated, and may play a role in epidemiological investigations, enrolment of individuals in survivor care programmes or research. Antibody detection from plasma or serum should not be used alone for diagnosis of cases. Serology relies on IgM detection in serum samples taken from recent acutely ill patients and/or IgG detection, in paired serum samples. Paired serum samples should be collected at least 21 days apart; ideally the first during the acute phase of illness and the second 3–4 weeks later to detect seroconversion. Recent vaccination may interfere with serology results. Interpretation of serological results is included in Table 5.

5.4 Detection of viral antigen by rapid diagnostic test

Antigen rapid diagnostic tests (Ag-RDTs) may be available for some *Orthoebolavirus* species. They can be beneficial tools when used for post-mortem diagnosis during an ongoing outbreak, to detect cases and reduce burden on teams carrying out safe and dignified burials. Ag-RDTs are not currently recommended to confirm new outbreaks of orthoebolavirus or orthomarburgvirus infections, including in countries where there is not an ongoing outbreak, as this should be done by NAAT. Current data show lower sensitivity of Ag-RDTs compared with NAAT, with a risk of missing EBOD cases. For some tests there is a lack of data on their performance outside laboratory conditions. The performance of Ag-RDTs should be validated prior to use against the relevant *Orthoebolavirus* species and orthomarburgviruses. Results of Ag-RDTs should be interpreted with caution, considering clinical and epidemiological factors. At the time of publication of this guidance, no performance data were available for use of Ag-RDTs to test for *Orthoebolavirus bundibugyoense.*

[3] The WHO Emergency Use Listing procedure, introduced in 2020, replaced the earlier Emergency Use Assessment and Listing mechanism developed during the 2014–2016 Ebola virus disease outbreak.

5.5 Genomic sequencing

Genome sequence data can provide valuable information to confirm a diagnosis, and support surveillance, outbreak investigation, and public health response. Genomic surveillance can support outbreak investigation by helping assess relatedness between cases, identify transmission chains, distinguish between single and multiple zoonotic spillover events, investigate cross-border transmission, and assess potential viral persistence in individuals who have recovered from infection. Genome sequence data are also important for monitoring viral evolution and for supporting the evaluation and ongoing performance of diagnostics, therapeutics and vaccines. Genomic sequencing may be used to enable species and virus identification (*Orthoebolavirus bundibugyoense, Orthoebolavirus sudanense, Orthoebolavirus taiense* or *Orthoebolavirus zairense,* MARV and RAVV).

Sequencing strategies should be defined according to the epidemiological context, public health objectives, laboratory capacity and available resources, with integration of genome sequence data into routine surveillance and outbreak investigation wherever feasible.

Several sequencing approaches may be used for genomic surveillance of orthoebolaviruses and orthomarburgviruses, including amplicon-based sequencing, targeted enrichment (probe capture or targeted metagenomic sequencing), and untargeted metagenomic sequencing. The selection of sequencing approaches should be guided by the target virus, intended public health objective, specimen type, expected viral load, available infrastructure and required genomic resolution.

Where validated schemes exist for the relevant virus species, amplicon-based sequencing approaches are generally the most sensitive and are well suited for routine genomic surveillance, including samples with lower viral loads or relatively high Cq/Ct values, although amplification efficiency may be affected by primer mismatches arising from viral genetic diversity. Targeted enrichment approaches may improve genome recovery and sequencing efficiency whereas untargeted metagenomic sequencing can support broader pathogen characterization, including identification of coinfections, divergent viruses or unexpected pathogens; however, it generally requires higher viral loads and has lower analytical sensitivity than targeted methods.

The ability to recover complete or near-complete viral genomes is influenced by specimen type, specimen quality, viral load, sequencing methodology and bioinformatic workflows. In general, samples with lower Cq/Ct values are more likely to yield high-quality genome coverage, whereas samples with higher Cq/Ct values may result in partial genome recovery or reduced sequencing sensitivity. Sequencing results should therefore be interpreted in conjunction with assay performance characteristics, epidemiological information and associated metadata.

WHO strongly encourages countries and laboratories to share genome sequence data and associated metadata, and where appropriate and feasible, raw sequence data, through publicly accessible databases in a timely manner to support global surveillance and risk assessment. Where feasible, consensus genome sequences should be accompanied by appropriate metadata and quality metrics to facilitate interpretation and downstream analyses.

5.6 Other techniques

Other advanced techniques can be used for phenotypic characterization, such as electron microscopy and viral culture. As these methods are not recommended as part of routine diagnosis, the specific details for these methodologies are not covered in this document.

Electron microscopy. Electron microscopy can be used to evaluate a sample for the presence of virions morphologically identical to orthoebolaviruses and orthomarburgviruses. With the availability of molecular assays and the high technical skills and facility required for electron microscopy, this method is not routinely used for the diagnosis of orthoebolaviruses and orthomarburgviruses.

Viral culture. Virus isolation is not recommended as a routine diagnostic procedure because it involves propagation of virus, requiring appropriate experience and maximum containment facilities, and takes days to perform.

6. Interpretation of laboratory results

Confirmation of orthoebolaviruses and orthomarburgviruses infections should consider clinical and epidemiological information. Positive detection using a pan-filovirus assay followed by confirmation of species via NAAT and/or sequencing, or positive detection using species specific NAAT in suspected cases, indicates confirmation of orthoebolaviruses and orthomarburgviruses infection.

Note that multiple factors could contribute to false-negative NAAT results, such as poor quality of sample, incorrect handling, shipping or storage, or technical reasons inherent to the test or related to test performance, *e.g.* RNA extraction failure, as well as nucleotide sequence diversity at the target genome region of the virus.

When the clinical presentation and epidemiology suggest an infection with EBOD or MVD, but NAAT results are negative, serological IgM and IgG testing may be useful to further investigate prior infection for epidemiological purposes or to facilitate enrolment in survivor care programmes. Serological results should always be interpreted with caution, considering the vaccination status of the patient.

Definitions of acute confirmed cases, confirmed convalescent cases, non-cases and equivocal cases according to laboratory results from EDTA-blood/plasma and serum testing are described in Table 5 below.

Table 5. Definitions of acute confirmed case, confirmed convalescent case, non-case and equivocal according to test results from EDTA-blood/plasma and serum, and presence of symptoms

Phase	Test results	Type of case
Symptomatic	NAAT (+)	Acute confirmed case
	NAAT (–) (after 72 hours following symptom onset)	Non-case
Post-symptomatic	NAAT (–) or not done, IgM (+) and IgG (+) or NAAT (–) or not done, IgM (–) and IgG (+) only if sampled more than 3 months after symptom onset and with clinical and epidemiologic findings compatible with an infection from an orthoebolavirus or an orthomarburgvirus.	Confirmed convalescent case
	NAAT (–) or not done during symptomatic phase, IgM (–) and IgG (+) only if tested less than 3 months since symptom onset and with clinical and epidemiology findings compatible with an infection from an orthoebolavirus or an orthomarburgvirus.	Equivocal

Because of the inverse relationship between Cq/Ct values and the amount of virus RNA in a patient sample, Cq/Ct values may serve as proxy for viral load estimates. Low Cq/Ct values suggest a high viral load and vice versa. However, many factors may impact the observed Cq/Ct value; examples of which are the type of sample taken, the sample quality, the conditions in which the sample was transported or stored, volumes used in different steps of the procedure, platforms and reagents used for extraction and RT-PCR, efficiency of the extraction and/or RT-PCR process, expertise of staff performing the test, validation processes, presence of inhibitors in the sample and potential contamination of the sample with disinfectant and other inhibitors. Therefore, Cq/Ct values should be interpreted with significant caution, should not be compared across different assays, platforms, laboratories or specimen types and should not be used as direct measures of infectiousness.

All test results should be interpreted in line with the manufacturer's recommendations.

7. Quality management

Standard operating procedures should be documented and maintained for all laboratory processes, including specimen receipt, testing, quality control, result validation, reporting, and corrective actions.

All reagents, control materials, and test kits should be stored according to the manufacturer's recommendations.

7.1 Quality control

Appropriate quality control measures should be implemented for all assays used, in accordance with good laboratory practice and the laboratory's quality management system. Quality control procedures should be tailored to the assay type and should monitor critical steps of the testing process, including specimen integrity, reagent performance, analytical accuracy, and result interpretation.

For NAATs, controls should be designed to monitor specimen integrity, nucleic acid extraction efficiency, amplification performance, and the presence of inhibitory substances. Positive and negative controls should be included in every assay run. Where not provided within commercial kits, suitable positive control materials may be obtained from specialized sources *(24)*. Positive controls should be included at a low but detectable concentration to adequately assess assay sensitivity and performance near the limit of detection. Given the high analytical sensitivity of NAATs, negative extraction controls and no-template controls should be included in each run to detect contamination introduced during specimen processing or assay setup. The use of extraction, amplification, internal, and inhibition controls can support the differentiation of true positive and true negative results from false-positive, false-negative, invalid, or inconclusive findings.

For serological assays, controls should monitor reagent integrity, assay performance, and consistency of interpretation. Where quantitative or semi-quantitative assays are used, calibration and verification procedures should be performed at defined intervals to ensure reproducibility and accuracy of results.

If any quality control criteria are not met, patient results should not be reported. The cause of the quality control failure should be investigated, corrective actions implemented, and testing repeated. All quality control activities, deviations, investigations, and corrective measures should be recorded and reviewed as part of the laboratory quality management system.

7.2 External quality assessment

Laboratories testing for *orthoebolaviruses* and *orthomarburgviruses*, at national and sub-national level, should participate in external quality assessment through proficiency testing, and/or retesting. It is recommended that for each outbreak of EBOD or MVD, the first samples tested (positive and negative) be sent to an international reference laboratory for filoviruses or a designated WHO Collaborating Centre (CC) for inter-laboratory comparison as part of external quality assurance implementation. The list of WHO CCs for VHF can be found in Annex 1.

8. Biological risk management

8.1 Biosafety requirements

Biosafety through the testing process

All samples collected should be regarded as potentially infectious and handled with extreme caution. Appropriate standard operating procedures must be in place and consistently followed. Personnel responsible for transporting and handling specimens must receive training and be assessed as competent in the safe and consistent performance of all assigned tasks before undertaking them. These may include donning and doffing and use and selection of personal protective equipment (PPE), sample collection, sample transportation, sample receipt, sample inactivation and extraction, performance of NAAT, serological testing, Ag-RDT, genomic sequencing or other tests, as applicable, waste management and disposal, spill decontamination procedures and full laboratory decontamination procedures in the event of closure or decommissioning of the facility.

Any minor wounds (e.g. cuts on the hands) should be covered before donning PPE. Major wounds should prohibit staff from handling samples. Laboratory personnel should wear appropriate PPE at all stages of the diagnostic process involving potential exposure to virus-containing material *(25)* and follow good microbiological practice and procedures.

It is recommended that all manipulations of samples originating from suspected or confirmed cases of EBOD or MVD be conducted according to a risk-based approach *(26)*. Each laboratory should conduct a local risk assessment for each process involved in handling samples. National guidelines on laboratory biosafety and biosecurity should be followed under all circumstances.

Heightened control measures are recommended in addition to the core requirements while handling samples that have not been inactivated *(26)*.

For the purpose of sample reception and clinical testing without virus propagation, these include the following:

- Personnel handling the reception of samples should wear appropriate additional PPEs. Additional PPEs should include additional body protection, respiratory protection, eye protection and footwear *(25)*. Samples should arrive at the laboratory within a triple package, each of them requiring disinfection using appropriate disinfectant, concentration and contact time before the tubes can be transferred for downstream processing (e.g. inactivation). If a patient sample has leaked into its secondary or tertiary layer of containment it should not be processed in the laboratory, and a new sample should be requested.

- Samples from patients with suspected EBOD or MVD should be handled in a certified Class III biosafety cabinet (or higher) or glovebox, to perform sample inactivation and aliquoting.[4]

[4] Handling of samples that were inactivated using a validated protocol does not require a biosafety cabinet or glovebox.

In the absence of a Class III biosafety cabinet or glovebox, a certified Class II biosafety cabinet may also be used with appropriate additional PPE *(25)* and training.

- The external surfaces of tubes containing inactivated samples should be decontaminated using an appropriate disinfectant and allowed the required contact time. Removal of tubes from the biosafety cabinet or glovebox should only occur once this disinfection has occurred.

- Sample containers and laboratory surfaces should be appropriately disinfected using an effective disinfectant at an appropriate concentration.

Additional control measures should be considered for specific procedures, for example for aerosol-generating procedures, according to the local risk assessment. For more information on biological risk management, see the *Laboratory biosafety manual, 4th edition (26)* particularly the monograph entitled *Outbreak preparedness and resilience (27)*.

Disinfectants

Effective disinfectants include freshly made 0.5% (or 5000 ppm) sodium hypochlorite solution and quaternary ammonium compounds. Note that for all use of disinfectants, an appropriate contact time should be observed (e.g., 10 minutes for 0.5%/5000 ppm sodium hypochlorite). After an appropriate contact time, 70% ethanol may be best used to wipe off sodium hypochlorite from working and instrument surfaces to avoid corrosion. Some disinfectants may interfere with NAAT, therefore care should be taken not to contaminate samples. Chlorine should not be used in the case of a spill of a sample mixed with lysis buffer, if the buffer contains guanidium thiocyanate as this would result in the forming of toxic hydrogen cyanide.

Management and disposal of waste

All waste that may contain orthoebolaviruses or orthomarburgviruses should be decontaminated at source before disposal by using an approved method, such as chemical disinfection and autoclaving, or chemical disinfection and incineration where an autoclave is not available. Sharps containers must be used for sharp items. All waste containers should be securely sealed and the outer surface decontaminated with appropriate disinfectant and contact time before disposal.

Items containing guanidium thiocyanate and chlorine should be kept separate as their mixing results in the forming of toxic hydrogen cyanide. Cartridges from automated or semi-automated testing platforms may require special conditions for disposal, for example segregated double bagging of used cartridges for incineration at high temperature to reduce risks associated with chemical components of the cartridges.

8.2 Biosecurity requirements

In addition to biosafety requirements, facilities performing testing for filoviruses should implement biosecurity measures consistent with the WHO *Laboratory biosecurity guidance (28)* to prevent unauthorized access, loss, theft, misuse or diversion of biological materials and associated information. Access to clinical specimens, reference materials and storage areas should be restricted to authorized personnel only, in accordance with national regulations. Laboratories should maintain accurate sample inventories and chain-of-custody records, ensure secure storage and controlled movement of materials, and establish procedures for the secure transport and shipment of specimens. Biosecurity awareness and regular staff training should be incorporated into laboratory quality and risk management programmes. Waste and residual materials containing potentially infectious biological material should be managed, transported and disposed of in a manner that minimizes opportunities for unauthorized access or release. These measures should be integrated throughout the specimen lifecycle, from collection and transport to storage, testing and final disposal.

8.3 Occupational health

All laboratory and transport personnel working with samples suspected or confirmed to contain orthoebolaviruses or orthomarburgviruses should immediately report any symptoms that meet the case definition to health authorities and their supervisor and should seek medical advice.

The physical and mental wellbeing of laboratory personnel is a critical component of laboratory biosafety management and should be supported to ensure that laboratory activities are conducted safely, consistently, and reliably. Working in a laboratory testing for EBOD or MVD can be a mentally and physically stressful environment. Incidents or accidents involving potential or actual exposure to samples containing orthoebolaviruses or orthomarburgviruses should also be immediately reported and any affected laboratory area/equipment appropriately decontaminated. Personnel who may have been exposed should seek medical advice as soon as possible. In case of laboratory exposure, medical countermeasures including vaccine administration may be considered, following national protocols.

National health authorities should conduct a risk assessment and consider whether, prior to or in response to an outbreak, immunization is required of health care workers, including laboratory personnel, who are at risk of exposure to individuals with or samples containing *Orthoebolavirus zairense*. For further recommendations on vaccine use, refer to the 2024 recommendations of the Strategic Advisory Group of Experts on Immunization *(29)*.

There are no licensed vaccines or therapeutics for pre- or post-exposure of laboratory personnel for other species of orthoebolaviruses, including *Orthoebolavirus bundibugyoense*, or for orthomarburgviruses, although research is being undertaken on therapeutics and vaccine candidates.

9. Reporting of cases and test results

9.1 National reporting requirements

Individual test results should be communicated promptly to the responsible health care provider, who should inform the patient or their family as soon as possible. Results should ideally be available within 24 to 48 hours of sample collection to enable the timely implementation of appropriate public health and infection prevention and control measures and clinical management if the case is confirmed. Results should include whether the sample was positive or negative, and the Cq/Ct value where applicable. It may be helpful to include a note in each report to support cautious interpretation of Cq/Ct values.

In the case of deceased individuals, the results should also be communicated to national authorities responsible for safe and dignified burials.

New positive cases should be reported to the public health/outbreak response team or national authorities in a timely manner, in order to facilitate contact tracing, case investigation and public health action.

In addition, information regarding the number of samples tested and the number of positive samples should be communicated by each testing laboratory to national authorities responsible for outbreak response and surveillance, according to national protocols.

An anonymized database should be established to collate and monitor laboratory testing. A line list of samples should also be in place, including the laboratory's unique identification number to facilitate a link back to the patients. It is also recommended that the following additional indicators are tracked weekly:

- average time from sample collection to result delivery
- average time from sample reception at the laboratory to result delivery
- the positivity rate among tested samples and by sample type
- the positivity rate among tested samples from new suspect cases and by sample type
- number of failed tests and reasons of failure (e.g. technical failures, procedural nonconformity, inhibition in samples)
- average consumption of test kits.

9.2 International reporting requirements

States Parties to the International Health Regulations (2005) are reminded of their obligations to notify WHO of events that may constitute a Public Health Emergency of International Concern, using the decision instrument in Annex 2 of the Regulations, and to continue to provide WHO with timely, accurate and sufficiently detailed public health information on notified events *(30)*.

10. Considerations for national testing strategies

10.1 Setting up a national strategy for testing

A national testing strategy for *Orthoebolavirus* and *Orthomarburgvirus* should be endorsed and implemented, as part of an integrated national laboratory system, to ensure clear organization of laboratory capacities in a country, and efficient sample transportation and laboratory information management. This can be undertaken while the country is not experiencing an active outbreak and updated as needed in the event of a confirmed outbreak, or an outbreak occurring in a neighbouring country.

Each country at risk of an outbreak should designate a national reference laboratory for Orthoebolavirus and Orthomarburgvirus testing within an integrated national laboratory system and in alignment with the broader capacity to detect epidemic-prone diseases. If *Orthoebolavirus* and/or *Orthomarburgvirus* testing is not available in-country, samples can be shipped internationally to a reference laboratory. The diagnostic strategy adopted by the national reference laboratory for *Orthoebolavirus* and *Orthomarburgvirus* detection should be based on testing objectives and a mapping of national testing capacities and resources. It should also include provisions for the storage, biobanking and/or safe destruction of samples (retention policy), in line with national quality, biosafety and biosecurity policies and international standards.

10.2 Adapting the national strategy to respond to a disease outbreak

A national testing strategy for the diagnosis of EBOD and MVD should be determined by available capacity, resources, and geographic context. The strategy should be informed by a risk assessment and adapted as the outbreak evolves. Key components should include the number, distribution, and capacity of laboratories; the availability of appropriate testing platforms; the number and competency of trained personnel; and the coverage and reliability of specimen transport systems to ensure timely delivery of samples to testing sites.

Centralized testing at a national reference laboratory using open NAAT platforms[5] offers advantages, including a controlled testing environment, greater flexibility, and the ability to perform differential diagnosis on site. However, where transport times to central laboratories are prolonged, countries should consider strengthening specimen referral and transport systems, which can also support broader pathogen testing, and/or introducing decentralized testing options.

[5] An open NAAT platform refers to the use particularly of a thermocycler which is open-source and non-proprietary.

Decentralized and surge testing approaches may be deployed in high-risk or highly affected areas to reduce turnaround times and rapidly increase testing capacity during an outbreak. These approaches are often complementary and may be implemented simultaneously. Approaches may involve expanding testing through the existing national and sub-national laboratory network, setting up field laboratories and/or use of validated near-patient or point-of-care NAAT devices in existing facilities or field laboratories.

Expanding and/or decentralizing capacity may involve scaling up existing national, regional, district, or peripheral laboratories to perform Orthoebolavirus and Orthomarburgvirus testing, based on the level of infrastructure, biosafety, and staff competency available. This may include upgrading equipment, strengthening biosafety and quality management systems, ensuring supply chains for reagents, and training and assessing staff competency. Where appropriate, near-patient or point-of-care NAAT testing may also be introduced.

In addition, field laboratories may be established through the deployment of national or international teams and the provision of mobile laboratory infrastructure and supplies, including through the deployment of rapid response mobile laboratories *(31)*.[6]

The choice of testing platform – whether open NAAT systems, semi-automated platforms, or near-patient devices – should be based on operational needs, available resources, and outbreak context.

More information on the specific steps to be taken when developing a strategy are described in Annex 2.

[6] Rapid response mobile laboratories are designed for national or international deployment to address laboratory and diagnostic gaps by providing surge capacity during emergencies and across all phases of the emergency management cycle. Rapid response mobile laboratories aim to deliver quality, needs-based critical diagnostic services in emergency settings through scalable and adaptable deployments tailored to specific mission objectives.

11. Global laboratory referral

WHO, through its regional and country offices, can assist Member States to access testing and inter-laboratory comparison through international referral of samples to international reference laboratories and WHO CCs for VHF. Where appropriate and safely performed, inactivation of samples in the local laboratory may facilitate referral and ease logistical challenges. Additional details on WHO CCs for VHF can be found in Annex 1.

References

1. Diagnostic testing for Ebola and Marburg virus diseases: interim guidance, 20 December 2024. Geneva: World Health Organization; 2024 (https://iris.who.int/handle/10665/380073). Licence: CC BY-NC-SA 3.0 IGO.

2. ICTV report on virus classification and taxon nomenclature, Family: Filoviridae, Genus: Orthoebolavirus [Website]. International Committee on Taxonomy of Viruses (https://ictv.global/report/chapter/filoviridae/filoviridae/orthoebolavirus, accessed 5 July 2026).

3. ICTV report on virus classification and taxon nomenclature, Family: Filoviridae, Genus: Orthomarburgvirus [Website]. International Committee on Taxonomy of Viruses. (https://ictv.global/report/chapter/filoviridae/filoviridae/orthomarburgvirus, accessed 5 July 2026).

4. ICTV report on virus classification and taxon nomenclature, Family: Filoviridae [Website]. International Committee on Taxonomy of Viruses (https://ictv.global/report/chapter/filoviridae/filoviridae, accessed 5 July 2026).

5. ICD-11, International Classification of Diseases 11th Revision, The global standard for diagnostic health information [Website]. International Committee on Taxonomy of Viruses. (https://icd.who.int/en, accessed 5 July 2026).

6. Rougeron V, Feldmann H, Grard G, Becker S, Leroy EM. Ebola and Marburg haemorrhagic fever. J Clin Virol. 2015 Mar;64:111-9 (https://doi.org/10.1016/j.jcv.2015.01.014).

7. Ebola virus outbreak toolbox, Updated June 2026 [Website]. Geneva: World Health Organization (https://www.who.int/emergencies/outbreak-toolkit/disease-outbreak-toolboxes/ebola-and-marburg-virus-outbreak-toolbox, accessed 6 July 2026).

8. WHO Fact sheet: Ebola disease, 24 April 2025 [Website]. Geneva: World Health Organization (https://www.who.int/news-room/fact-sheets/detail/ebola-virus-disease, accessed 6 July 2026).

9. WHO Fact sheet: Marburg virus disease, 20 January 2025 [Website]. Geneva: World Health Organization (https://www.who.int/news-room/fact-sheets/detail/marburg-virus-disease, accessed 6 July 2026).

10. Optimized supportive care for Ebola virus disease: clinical management standard operating procedures. Geneva: World Health Organization; 2019. (https://iris.who.int/handle/10665/325000). Licence: CC BY-NC-SA 3.0 IGO.

11. Therapeutics for Ebola virus disease, 19 August 2022. Geneva: World Health Organization; 2022. (https://iris.who.int/handle/10665/361697). Licence: CC BY-NC-SA 3.0 IGO.

12. Bào Y, Amarasinghe GK, Basler CF, Bavari S, Bukreyev A, Chandran K, Dolnik O, Dye JM, Ebihara H, Formenty P, Hewson R, Kobinger GP, Leroy EM, Mühlberger E, Netesov SV, Patterson JL, Paweska JT, Smither SJ, Takada A, Towner JS, Volchkov VE, Wahl-Jensen V, Kuhn JH. Implementation of Objective PASC-Derived Taxon Demarcation Criteria for Official Classification of Filoviruses. Viruses. 2017 May 11;9(5):106 (https://doi.org/10.3390/v9050106).

13. Burk R, Bollinger L, Johnson JC, Wada J, Radoshitzky SR, Palacios G, Bavari S, Jahrling PB, Kuhn JH. Neglected filoviruses. FEMS Microbiol Rev. 2016 Jul;40(4):494-519 (https://doi.org/10.1093/femsre/fuw010).

14. WHO guidelines for the clinical management of filovirus disease. Geneva: World Health Organization; 2026 (https://iris.who.int/handle/10665/386380). Licence: CC BY-NC-SA 3.0 IGO.

15. Strecker T, Palyi B, Ellerbrok H, Jonckheere S, de Clerck H, Bore JA, Gabriel M, Stoecker K, Eickmann M, van Herp M, Formenty P, Di Caro A, Becker S. Field Evaluation of Capillary Blood Samples as a Collection Specimen for the Rapid Diagnosis of Ebola Virus Infection During an Outbreak Emergency. Clin Infect Dis. 2015 Sep 1;61(5):669-75 (https://doi.org/10.1093/cid/civ397).

16. How to safely collect blood samples by phlebotomy from patients suspected to be infected with Ebola: interim guidance. Geneva: World Health Organization; 2014 (https://iris.who.int/handle/10665/144818).

17. Interim guideline: how to safely collect oral swabs (saliva) from deceased patients suspected to be infected with Ebola. Geneva: World Health Organization; 2014 (https://iris.who.int/handle/10665/136556).

18. Palyi B, Magyar N, Henczko J, Szalai B, Farkas A, Strecker T, Takacs M, Kis Z. Determining the effect of different environmental conditions on Ebola virus viability in clinically relevant specimens. Emerg Microbes Infect. 2018 Mar 29;7(1):52 (https://doi.org/10.1038/s41426-018-0043-z).

19. UN Model Regulations Rev. 23 (2023) [Website]. United Nations Economic Commission for Europe: Geneva (https://unece.org/transport/dangerous-goods/un-model-regulations-rev-23, accessed 6 July 2026).

20. Guidance on regulations for the transport of infectious substances 2025–2026: applicable as from 1 October 2025. Geneva: World Health Organization; 2026 (https://iris.who.int/handle/10665/385528). Licence: CC BY-NC-SA 3.0 IGO.

21. WHO Prequalification of Medical Products, Ebola virus disease, Ebola IVD EUAL status after PHEIC termination [Website]. Geneva: World Health Organization (https://extranet.who.int/prequal/vitro-diagnostics/ebola-virus-disease, accessed 6 July 2026).

22. Emergency use listing procedure, version 9 August 2022. Geneva: World Health Organization; 2022 (https://www.who.int/publications/m/item/emergency-use-listing-procedure).

23. WHO Prequalification of Medical Products, Bundibugyo virus disease PHEIC – Emergency Use Listing Procedure (EUL) for IVDs [Website]. Geneva: World Health Organization (https://extranet.who.int/prequal/vitro-diagnostics/bundibugyo-virus-disease-pheic-emergency-use-listing-procedure-eul-ivds, accessed 6 July 2026).

24. European Virus Archive [Website]. (https://www.european-virus-archive.com/, accessed 6 July 2026).

25. Personal protective equipment. Geneva: World Health Organization; 2020 (https://iris.who.int/handle/10665/337961). Licence: CC BY-NC SA 3.0 IGO.

26. Laboratory biosafety manual, 4th ed. Geneva: World Health Organization; 2020 (https://iris.who.int/handle/10665/337956). Licence: CC BY-NC-SA 3.0 IGO.

27. Outbreak preparedness and resilience. Geneva: World Health Organization; 2020 (https://iris.who.int/handle/10665/337959). Licence: CC BY-NC-SA 3.0 IGO.

28. Laboratory biosecurity guidance. Geneva: World Health Organization; 2024 (https://iris.who.int/handle/10665/377754). Licence: CC BY-NC-SA 3.0 IGO.

29. Extraordinary meeting of the Strategic Advisory Group of Experts on Immunization on Ebola vaccination, May 2024: conclusions and recommendations *Weekly Epidemiological Record, 99* (27), 355–362. World Health Organization; 2024. (https://iris.who.int/handle/10665/378109).

30. International Health Regulations (2005), as amended in 2014, 2022 and 2024. Geneva: World Health Organization (https://apps.who.int/gb/bd/pdf_files/IHR_2014-2022-2024-en.pdf).

31. Guidance for rapid response mobile laboratory (RRML) classification. World Health Organization: Regional Office for Europe; 2021 (https://iris.who.int/handle/10665/339845). Licence: CC BY-NC-SA 3.0 IGO.

Annex 1
List of WHO CCs for VHF

1. WHO Collaborating Centre for Arboviruses and Viral Haemorrhagic Fevers, Centre International de Recherches Médicales de Franceville, Gabon

2. WHO Collaborating Centre for Arbovirus and Haemorrhagic Fever Reference and Research, Bernhard-Nocht Institute for Tropical Medicine, Germany

3. WHO Collaborating Centre for Arboviral and Zoonotic diseases preparedness research and reference, Erasmus University Hospital, Netherlands (Kingdom of the)

4. Centre collaborateur de l'OMS pour les Arbovirus et les Virus de Fièvres Hémorragiques, Institut Pasteur de Dakar, Senegal

5. WHO Collaborating Centre for Virus Reference & Research (Special Pathogens), United Kingdom Health Security Agency, United Kingdom of Great Britain and Northern Ireland

6. WHO Collaborating Centre for Viral Hemorrhagic Fevers, Centers for Disease Control and Prevention, United States of America

Annex 2
National testing strategies for EBOD and MVD diagnosis

The following steps can be considered when developing the national laboratory strategy for EBOD and MVD diagnosis. Ideally, this should be done before there is an active outbreak.

Endorsing a national *Orthoebolavirus* and *Orthomarburgvirus* testing strategy and standard operating procedures

Step 1. Map and assess laboratory resources (equipment, reagents and consumables) in the country: laboratory testing sites, sample referral and transport routes, equipment for sample inactivation, nucleic acid extraction and amplification (e.g. glovebox, RNA extraction kits, thermocyclers, semi-automated platforms, RT-PCR kits, and other reagents and consumables, etc.).

Step 2. Designate or confirm the national reference laboratory for orthoebolaviruses and orthomarburgviruses and ensure that access to an international reference laboratory or WHO CC is available for inter-laboratory comparison and/or further characterization.

Step 3. Develop and endorse a national testing strategy based on NAAT for orthoebolaviruses and orthomarburgviruses and share with all relevant laboratories in country.

Step 4. Develop or update and adopt national standard operating procedures for sample collection, handling, transport, reception, processing and testing, data management and data sharing and distribute to all involved laboratories for implementation. Provide training on the standard operating procedures to relevant staff if necessary and assess competencies.

Preparing for testing

Step 5. Assess the capabilities and requirements of each laboratory with regard to number of samples to test per week; reagents, equipment (e.g. glovebox) and consumables required for the number of samples.

Step 6. Assess sample transport capacities in country and internationally, considering:

- possible routes of transportation
- availability of vehicles, drivers and fuel and training of drivers on biosafety
- resources required, e.g., Category A shipping material
- availability of certified shippers (or the need for training)
- designated VHF reference laboratory, such as a WHO CC, for sample referral and confirmatory testing
- the need to have a Material Transfer Agreement ready.

Step 7. Assess training needs of laboratory workforce and consider carrying out simulation exercises to test readiness for outbreak response.

Step 8. Purchase any additional equipment, reagents and consumables needed and implement any trainings and competency testing required. Requests can be made to WHO and/or other partner organizations for technical support.

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